Entra ID Identity Governance Lab

Nov 5, 2026 · @Omar

Overview

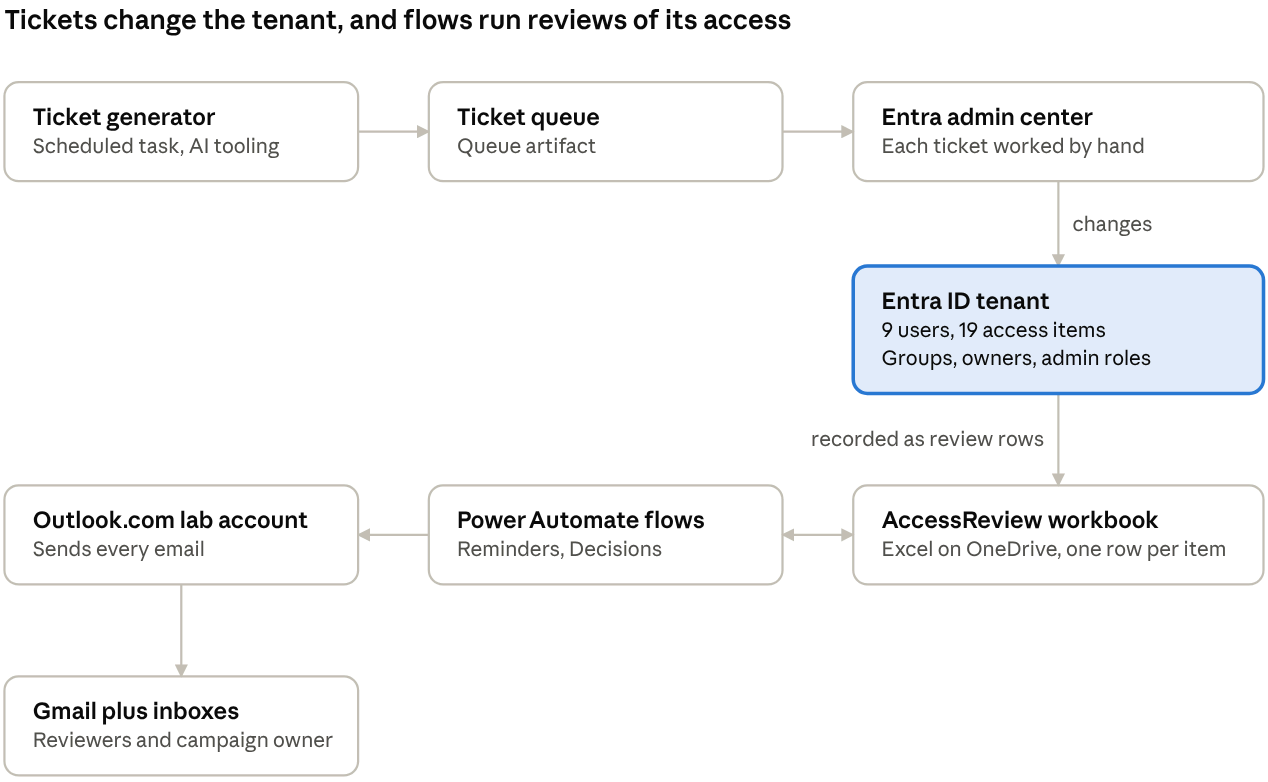
This lab is a personal Microsoft Entra ID tenant where I practice the identity and access work an IAM analyst does, from access requests to access review campaigns. It runs on a self directed tenant with sample users, not a production environment.

The lab has two parts.

1. A practice IAM help desk. AI assisted tooling generates realistic identity and access tickets, and I work each one by hand in the Entra admin center.
2. An automated access review campaign. Three Power Automate flows send reviewer reminders, escalate overdue reviews, and record each Keep or Remove decision with a date as audit evidence.

The lab builds on three years as an Applications Analyst at OceanFirst Bank, where I handled user access requests under least privilege principles and managed production changes through formal change control.

Architecture



lab architecture · help desk and review automation

The top row is the help desk, where tickets drive changes in the tenant. The bottom rows are the review campaign, where flows read and update the review workbook and email reviewers through Outlook.com.

Tenant and accounts

The tenant is osentra001gmail.onmicrosoft.com, with no paid Microsoft 365 licenses assigned. Each account below has one job, which keeps admin work, personal sign ins, and email separate.

Account or service	Type	Used for
omar singer (osentra001@gmail.com, shows as #EXT#)	Personal Microsoft account linked into the tenant as a Member	Daily admin work in the Entra admin center

Account or service	Type	Used for
Omar Admin (omar.admin@osentra001gmail.onmicrosoft.com)	Cloud account created in the tenant, Global Administrator	Power Automate and Power Platform sign in
osentra001@gmail.com on OneDrive	Personal Microsoft account	Holds the AccessReview Excel workbook
Outlook.com lab address	Personal Microsoft account	Sends all flow emails
Gmail plus addresses (osentra001+name@gmail.com)	Aliases of one Gmail inbox	Stand in for reviewer inboxes

Power Platform environment	Type	Status
Default Directory	Default	Available
Omar Lab	Developer	Holds the three flows. Dataverse could not be added without a license.

Identity inventory

Nine DC Comics themed users hold 19 pieces of access across four department groups, group ownership, and admin roles. This table matches the AccessReview sheet, where each row is one user and one access item.

User	Access items	Reviewer
Barbara Gordon	IT Group, Authen Admin, Group Admin, Guest Inviter, Helpdesk Admin	Victor Stone
Barry Allen	Finance Group	Bruce Wayne
Bruce Wayne	Finance Owner, Fin Auditor Owner	Omar Singer
Cassandra Cain	HR Group	Dick Grayson

User	Access items	Reviewer
Dick Grayson	HR Owner, HR Group	Omar Singer
Justin White	Marketing Group, Marketing Owner	Omar Singer
Kara Olsen	Marketing Group	Justin White
Selina Kyle	Finance Group, Fin Auditor Owner	Bruce Wayne
Victor Stone	IT Owner, IT Group, User Admin	Omar Singer

Reviewers follow two rules. Group owners review the people in their groups, and Omar Singer acts as the fallback reviewer for the owners at the top. Nobody reviews their own access.

Barbara Gordon holds four admin roles, which is the kind of excess access a review should question. In the first campaign her Guest Inviter role was marked Remove, and Victor Stone's IT Owner access was marked Keep.

Practice help desk

A scheduled task generates realistic identity and access tickets and files them into a queue, and I work each ticket by hand in the Entra admin center. Working requests from a queue gives the lab the steady flow of an operations team instead of one off exercises.

1. A scheduled task uses AI assisted tooling to write new tickets with the lab's users, groups, and roles.
2. The tickets land in a queue artifact.
3. I pick up a ticket and make the change in the live admin center.

Ticket types covered so far:

- Provisioning new users
- Deprovisioning users who leave
- Group membership changes
- Access certifications
- Lifecycle state changes

Access review automation

Three Power Automate flows run an access review campaign from one Excel table, and all three were tested end to end on the 19 row campaign. The table lives in the AccessReview workbook on OneDrive, in a table named Review.

Column	Purpose
ReviewID	Unique key the flows use to find and update a row
User, AccessItem	Who has what
ReviewerName, ReviewerEmail	Who decides, and where the reminder goes
DueDate	When the decision is due
Decision	Pending, Keep, or Remove
DecisionDate	Stamped by the flow once a decision is made
ReminderCount	How many reminders went out for that row

Flow	Schedule	What it does
Access Review Reminders	Daily, kept off or weekly between tests	Reads Pending rows. Emails an escalation to the campaign owner when a review is overdue, emails a reminder to the reviewer when it is not, and adds 1 to ReminderCount.
Access Review Decisions	Daily	Finds rows with a Keep or Remove decision and no date. Stamps today's date and emails a confirmation as evidence.
Test Email	Manual	One action that confirmed the account could run flows and send email

Build choices that made the flows reliable:

- Apply to each runs with concurrency set to 1, so Excel updates one row at a time and never locks.

- Values from a Filter array are read with expressions such as items('Apply_to_each')?['ReviewID'], added through the fx button.
- Dates come back in ISO 8601 format so the overdue check compares cleanly against utcNow().
- Emails go out through the Outlook.com connector from a dedicated lab address.

Test results: the escalation path fired for the 3 overdue rows, reminders went to the other 14 Pending rows, and the Keep and Remove rows were stamped with a decision date.

Problems solved

Most of the build time went to account types, licensing, and connector limits, not the flow logic. Each fix below came from reading the error, finding the cause, and changing the design.

Problem	Cause	Fix
Power Automate sign in rejected the Gmail account	It is a personal Microsoft account, and Power Platform needs a work account in the tenant	Created a cloud account in the tenant with Global Administrator
The existing #EXT# member account also failed	#EXT# means the account still signs in through the personal Gmail identity	Kept it for Entra admin work and used the new cloud account for Power Platform
Dataverse database would not create	The tenant has no license, and the Developer Plan did not assign one	Moved the review table to Excel on OneDrive
Excel connector refused the work account	Excel Online (OneDrive) only supports personal accounts	Connected it with the personal Microsoft account that owns the workbook
Email action returned Unauthorized	Microsoft restricts the built in Mail connector for new tenants	Sent email through the Outlook.com connector from a new lab address
Outlook connection failed with a work account error	The Office 365 Outlook connector was picked by mistake	Picked Send an email (V2) under Outlook.com instead

Problem	Cause	Fix
A new run started every minute	The Recurrence trigger was set to minutes	Turned the flow off, cancelled the runs, set it to 1 Day
Runs hung for over an hour	Parallel loop iterations locked the Excel file	Set Apply to each concurrency to 1 and kept the workbook closed during runs
Excel cells showed expression text	Expressions were typed into fields as plain text	Re-added them through the fx button so they evaluate
Update a row could not find rows	Same issue in Key Value	Re-added the Key Value expression through fx
Decision flow processed all 19 rows	Placeholder values like No Change counted as decisions	Standardized Decision to Pending, Keep, or Remove

Skills shown and talking points

The lab shows hands on IAM operations and access governance automation, the two areas most junior IAM and recertification postings ask for.

Skill	Where it shows up
User lifecycle (provisioning, deprovisioning, lifecycle changes)	Practice help desk tickets
Group membership and role management	Help desk tickets and the identity inventory
Least privilege and excess access review	Barbara Gordon's admin roles flagged and one removed
Access certification campaigns	The 19 row review campaign
Reviewer assignment and fallback reviewers	Group owners review members, Omar Singer covers owners
Audit evidence	Decision dates and confirmation emails

Skill	Where it shows up
Power Automate (personal lab)	Three working flows with loops, conditions, expressions, and Excel and Outlook.com connectors
Troubleshooting	The Problems solved table

Resume bullets that match this lab:

> Built a simulated identity governance environment using a personal Microsoft Entra ID tenant to practice real access control operations. This includes provisioning, deprovisioning, group membership changes, access certifications, and lifecycle state management with all changes applied in the live portal.

> Built AI assisted tooling to generate realistic identity and access tickets, giving me a more consistent simulation to work through and building familiarity with automated governance workflows.

> Built Power Automate workflows that track access review decisions in Excel, send reviewer reminders, escalate overdue reviews, and stamp each decision with a date as audit evidence.

Interview talking points:

- **Why build it.** I am building toward an information security career, so I built a tenant where I work identity requests and run a review campaign end to end.
- **How reviews work.** Each row is one user and one access item. The group owner decides Keep or Remove, and overdue reviews escalate to the campaign owner.
- **How decisions are proven.** Each decision gets a date stamp on its row and a confirmation email, which is the evidence an auditor asks for.
- **A hard problem.** The built in email connector was blocked for new tenants, so I traced the Unauthorized error to that restriction and switched to Outlook.com.
- **Be clear it is a lab.** Say plainly that the tenant is a personal lab with sample users, not production.

Next steps and open questions

The core lab is done. These additions would each cover another common job requirement.

- [ ] Change the reminder flow to send one summary email per run, built with Create HTML table
- [ ] Add a weekly status report with Pending, Keep, and Remove counts and a completion rate
- [ ] Build a Power BI Desktop dashboard on the AccessReview workbook
- [ ] Remove a user from the Entra group automatically when a reviewer picks Remove, if the connectors work without a paid license
- [ ] Set managers and group owners in Entra so reviewer assignments match the directory
- [ ] Publish this write up on the planned GitHub Pages portfolio alongside the SOC home lab

Open questions to confirm before publishing:

- Exact names of the admin roles shown in the sheet as Authen Admin, Group Admin, Helpdesk Admin, and User Admin
- Whether the group owners in the sheet are set as owners in Entra or only recorded in the sheet
- Which ticket types have been worked the most, with counts if the queue tracks them